Swapnil Saha

Contact

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Relevant Links

- **LinkedIn:** https://www.linkedin.com/in/swapnil-saha-b5aa2a37a/
- **GitHub:** https://github.com/SwapnilSaha5768

Skills

Programming Languages:

Python, C++, JavaScript, HTML, CSS, MySQL, RISC-V

Frameworks & Tools:

React, Node.js, Express, MongoDB, OpenGL, Arduino

Graphics Designing:

Web Design, Canva

Soft Skills:

- Teamwork, Project Management
- Leadership
- Problem Solving, Critical thinking
- Time Management

Other Skills:

MS Office (Word, Excel, PowerPoint), LaTeX

Objective

Aspiring software developer and researcher passionate about coding, problem-solving, and AI innovation. Combines strong technical foundation with leadership experience from organizing university events and leading creative teams. Seeking opportunities to contribute to impactful software projects, graphics designing and pursue advanced research in computer science.

Education

BRAC UNIVERSITY
Bachelor of Science in Computer Science & Engineering
Major: CSE
CGPA: 3.39 **(2021 -2025)**

DHAKA RESIDENTIAL MODEL COLLEGE
Higher Secondary Certificate
GPA: 5.00 **(2018 -2020)**

DHAKA RESIDENTIAL MODEL COLLEGE
Secondary School Certificate
GPA: 5.00 **(2010 -2018)**

RESEARCH

DEEPFAKE SPEECH RECOGNITION: EVALUATING ACCURACY AND EFFICIENCY IN DETECTION ALGORITHMS

The thesis investigates the challenges associated with AI-driven voice impersonation and evaluates techniques for detecting synthetic speech. It analyzes existing deepfake speech recognition models in terms of their accuracy and computational efficiency, identifies shortcomings, and suggests effective countermeasures to ensure secure audio communication.

Project

- **E-COMMERCE WEBSITE (MERN STACK)**
  Developed a full stack online shopping platform named Dalbhaat featuring user login, product browsing, cart functionality, checkout etc. Included an admin dashboard for inventory management and order tracking. Focus on secure form handling and efficient database integration.

- **UNIVERSITY BLOGGER SITE (PHP, MYSQL)**
  This blogging platform is built with PHP and MySQL, providing a robust backend for user registration, login, and content management. Users can create, edit, and publish blog posts through an easy-to-use interface and manage their profiles seamlessly. The system securely stores user and post data in the MySQL database, ensuring reliable performance and data integrity.

- **DISCORD MODERATION BOT (JAVASCRIPT)**
  This Discord bot is designed to enhance server management and entertainment by combining powerful moderation tools with music playback capabilities. It offers features like automatic message filtering, user warnings, and role management to keep the community safe and organized. Additionally, the bot supports playing music from popular sources, allowing users to queue, skip, and control tracks seamlessly within voice channels.

- **MAZE KEY QUEST (OPENGL/PYTHON)**
  Developed a 2D game where players navigate through a maze to collect points within a limited time frame. The objective is to solve the maze efficiently while avoiding moving balls touching any results in an immediate game over. The game challenges players' problem-solving and reflex skills under time pressure.

- **SLOT MACHINE (ASSEMBLY LANGUAGE)**
  Developed a text-based Slot Machine game in Assembly language using DOS interrupts. Implemented core features including random number generation, multiple win conditions (jackpot, sequences, matches), credit management, and user interaction through keyboard input. The program demonstrates proficiency in low-level programming, control flow, and display handling in a resource-constrained environment.

- **MAZE SOLVING ROBOT (DIGITAL LOGIC)**
  Built a 4-wheeler autonomous robot equipped with sensors to navigate and solve mazes, finding the safe exit without human intervention. Implemented real-time obstacle detection and pathfinding algorithms to guide movement. Demonstrated practical application of embedded systems and sensor integration for robotic control.

- **SMART HELMET (C++)**
  Designed and implemented a Smart Guardian Helmet with helmet detection, alcohol and drowsiness sensing, environmental alerts, theft prevention, and auto light control. Used Arduino-based sensor integration and RF wireless communication for real-time monitoring and engine control. Demonstrated embedded systems and IoT safety applications.

- **UNCERTAINTY-AWARE DEEP CLUSTERING WITH AUTOENCODER (PYTHON)**
  Developed a deep clustering pipeline using convolutional autoencoders and DEC (Deep Embedded Clustering) for unsupervised image classification. Compared Bayesian (MC-Dropout) and deterministic encoders on real-world image data (cats vs. dogs), evaluating clustering quality and model uncertainty using ARI, NMI, Silhouette Score, and stability analysis.

Leadership & EXTRACURRICULARS

- **FORMER VICE PRESIDENT IN BUSINESS DEVELOPMENT (IABC | BRAC UNIVERSITY)**
  Served as Vice President of Business Development at IABC BRAC Club, where I led key initiatives including organizing major events such as Insight, Influx, and Diplomat's Meet. Actively contributed to strategic planning, partnership development, and team coordination. Mentored and guided junior members, fostering leadership and collaboration within the club.

- **HEAD OF DIRECTORS IN FOOTBALL CLUB OF BRAC UNIVERSITY**
  Currently serving as Head of Directors at FCBU Club, after over two years of dedicated contribution in graphics design and social media management. Successfully handled the club's official Facebook and Instagram pages, creating engaging content and maintaining brand consistency. Recognized and promoted for outstanding performance, creativity, and commitment to the club's growth and outreach.

- **ORGANIZER, TEDXBRACU (2025)**
  Organizer of TEDxBRACU — led end-to-end planning of a TED-standard event, from acquiring the license to curating the theme, building and managing a multidisciplinary team, and handling logistics and compliance. Hosted a distinguished lineup of speakers from across industries. This role sharpened my skills in strategic leadership, stakeholder management, and public engagement.

- **FORMER SENIOR EXECUTIVE IN BRAC UNIVERSITY CHESS CLUB**
  Worked in HR Department at BRAC University Chess Club. Contributed to sponsorship efforts by drafting proposal letters and assisted in organizing events. Also responsible for managing and maintaining the club's member database.

- **VOULUNTEER, REMIAN LANGUAGE CLUB (2019)**
  Volunteering at the 6th Remain Language Club event, I helped coordinate activities, managed logistics, and supported facilitators to ensure everything ran smoothly. This role strengthened my teamwork and organizational skills while contributing to a positive language learning experience for all participants

Language

- English (Fluent)
- Bangla (Native)
- Hindi (Conversational)